MIL-R-27384B

28 October 1970

SUPERSEDING
MIL-R-27384A
30 June 1965

MILITARY SPECIFICATION

RIVET, BLIND, DRIVE TYPE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers drive pin type blind rivets which can be installed in assemblies or construction where access to only one side is available (see 6.1).

1.2 Classification.

1.2.1 Types and grades. Rivets covered by this specification shall be of the following types and grades, as specified in the invitation for bids (see 6.2).

Type I - Universal head
  Grade A - Aluminum-alloy bodies
  Grade B - Steel bodies

Type II - Countersunk head
  Grade A - Aluminum-alloy bodies
  Grade B - Steel bodies

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

FSC 5320

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SPECIFICATIONS

Federal

QQ-A-430 - Aluminum Alloy Rod and Wire; for Rivets and Cold Heading.
QQ-P-35 - Passivation Treatments for Austenitic, Ferritic and Martensitic Corrosion-Resisting Steel (Fastening Devices).
QQ-P-416 - Plating, Cadmium (Electrodeposited).

Military

MIL-H-3982 - Hardware (Fasteners and Related Items); Packaging and Packing for Shipment and Storage of.
MIL-C-5541 - Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys.
MIL-H-6088 - Heat Treatment of Aluminum Alloys.
MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys.

STANDARDS

Federal

Fed Std No. 66 - Steel: Chemical Composition and Hardenability.
Fed Test Method Std No. 151 - Metals; Test Methods.

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1312 - Fasteners, Test Methods.
MS24661 - Rivet, Blind - Countersunk Head, Drive Type.
MS24662 - Rivet, Blind - Universal Head, Drive Type.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Rivet bodies. Grade A rivet bodies shall be made of aluminum-alloy wire conforming to QQ-A-430, Composition 2117-T4 or Composition 5056-H32 (see 6.2). Grade B rivet bodies shall be made from carbon steel wire conforming to the requirements of Fed Std No. 66, AISI Numbers C1006 thru C1041 (see 6.2).

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3.1.2 Drive pins. Drive pins for Grade A, Composition 2117-T4 rivet bodies, shall conform to Fed. Std. No. 66, CRES 300 series; for Grade A, Composition 5056-H32 rivet bodies shall conform to Fed. Std. No. 66, CRES 305 only. Drive pins for Grade B rivet bodies shall conform to Fed. Std. No. 66, CRES 300 series or carbon steel.

3.2 Protective finish.

3.2.1 Unless otherwise specified in the invitations for bid, contract or order (see 6.2) rivet bodies and drive pins shall be plated, coated, or treated in accordance with 3.2.2 through 3.2.4.

3.2.2 Anodizing or chemical treatment. Grade A rivet bodies, Composition 2117-T4 aluminum alloy, shall be anodized in accordance with MIL-A-8625 or chemically surface treated in accordance with MIL-C-5541. Anodizing of Composition 5056-H32 aluminum alloy rivet bodies is not necessary.

3.2.3 Cadmium-plating. Carbon steel pins and Grade B rivet bodies shall be cadmium-plated. Cadmium-plating shall conform to the requirements of QQ-P-416, Type II, class 2.

3.2.4 Passivation. The corrosion-resisting steel drive pins shall be passivated in accordance with QQ-P-35.

3.3 Heat treatment.

3.3.1 Grade A (aluminum alloy) rivets. Heat treatment of 2117-T4 rivets shall conform to MIL-H-6088; Composition 5056-H32 rivets shall conform to the requirements of QQ-A-430.

3.3.2 Grade B (steel) rivets. The hardness of Grade B rivet bodies shall not exceed Rockwell B-95.

3.4 Design and dimensions. Design, dimensions, and tolerances shall be in accordance with MS24661 or MS24662, as applicable. Rivets shall conform to dimensions and tolerances after application of protective finish.

3.5 Identification.

3.5.1 Trade mark. Trade marks or other markings may be applied to the rivets provided they do not interfere with the dimensions, proper use of the rivets, or produce stresses that may result in failure.

3.6 Workmanship. Rivets shall be free from seams, fins, clinch or die marks and corrosion.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as

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otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot, for inspection purposes, shall consist of rivets that are of the same material, type, finish, size, and made under essentially the same conditions and submitted for acceptance at one time.

4.3 Sampling for lot acceptance.

4.3.1 Sampling for examination. A random sample of rivets shall be taken from each lot in accordance with MIL-STD-105 at Inspection Level II. The Acceptable Quality Level (AQL) shall be made in accordance with Table I herein.

4.3.2 Sampling for tests. A random sample of rivets shall be taken from each lot in accordance with MIL-STD-105, Inspection Level S-3. The AQL shall be 2.5 percent defective. A sample piece of aluminum alloy wire processed along with each heat treat lot of Grade A rivets may be used for the tensile test of 4.5.3.

4.3.3 Sampling for packaging and packing. Sampling for inspection and tests of packaging and packing shall be in accordance with MIL-H-3982.

4.4 Examination.

4.4.1 Visual and dimensional. Sample pins and rivets shall be examined to verify compliance with this specification in accordance with the following classification of defects and MIL-STD-105.

TABLE I. Classification of Defects

Categories	Defects	Method of Inspection
Critical	None	
Major	AQL = 1.5 percent	
101	Rivet diameter (3.4)	Measure
102	Angle of countersunk head (3.4)	Measure
103	Rivet length (3.4)	Measure
Minor	AQL = 6.5 percent	
201	Protective finish (3.2)	Visual
202	Head height (3.4)	Measure
203	Head diameter (3.4)	Measure
204	Overall length (3.4)	Measure
205	Workmanship (3.6)	Visual

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4.4.2 Protective coating. Protective coating shall be checked as indicated on the applicable MS standard.

4.4.3 Examination and tests for packaging and packing. Packaging and packing of the rivets, as specified in Section 5 herein, shall be inspected in accordance with MIL-H-3982.

4.5 Tests.

4.5.1 Chemical analysis. Sample rivet bodies and pins shall be tested for chemical composition of their respective materials in accordance with Method 111.2 and 112.2 of Fed Test Method Std. No. 151. Material manufacturer's certification may be accepted in lieu of this test.

4.5.2 Hardness test of Grade B rivets. Sample Grade B rivets shall be tested for the hardness requirements in accordance with MIL-STD-1312, Test No. 6 (3.3.2 and 6.2).

4.5.3 Tensile test of Grade A rivet material. The sample wire of Grade A rivet material (4.3.2) shall be tested for tensile properties in accordance with MIL-STD-1312, Test No. 8.

4.5.4 Protective coating. Protective coating and plating tests shall be conducted in accordance with the applicable specifications shown in 3.2.2 thru 3.2.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking shall be in accordance with MIL-H-3982.

5.2 Levels. Levels of packaging and packing shall be as specified by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. Rivets covered by this specification are intended for riveting structures or equipment whereon small rivets can be used and where only one side is accessible. Special tools are not required for the installation of these rivets.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Title and number of applicable Military standard.
c. Type and grade of rivets required (see 1.2.1).

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d. Choice of aluminum composition numbers and steel Fed Std No. 66 AISI numbers (see 3.1.1 and 3.1.2).
e. Protective finish (see 3.2).
f. Hardness test (3.3.2 and 4.5.2).
g. If certificate of conformance is acceptable for chemical analysis (4.5.1).
h. Level of packaging and packing (see 5.2).

Custodians:
Army - WC
Navy - YD
Air Force - 82

Review activities:
Army - AT, AV, ME, MI, MU
Navy - None
Air Force - 85
DSA-IS
NSA

User activities:
Army - EL, GL
Navy - AS, MC, SH
Air Force - None

Preparing activity:
Army - WC

Project number:
5320-0179

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